Irene Huang

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Education

Stony Brook University — Stony Brook, NY
M.B.A., General — *Expected Dec 2026*
GPA: 3.97/4.00

B.S., Computer Science — *May 2025*
GPA: 3.65/4.00

Research Interests

Electricity market design and co-optimized dispatch; stochastic optimization for storage dispatch under renewable and demand uncertainty; multi-day optimization for grid operations; energy storage systems; energy policy implementation.

Research Experience

Text-Informed Nutrition Prioritization for West Africa [GitHub] — 2026
Independent Research Project

- Developed a Python-based framework to combine West Africa Food Composition Table data with PubMed deficiency literature and prioritize foods in resource-constrained settings. Excluded foods with missing nutrient values, producing a smaller, but more complete dataset.
- Mined PubMed deficiency-related research for nutrient frequencies, converted them to weights, and applied TOPSIS to rank foods by deficiency relevance. Identified iron as the strongest deficiency signal and visualized results through heatmaps and comparative plots.
- Framed as exploratory, since local food availability, affordability, and consumption data were not available to assess practical relevance.

Data Envelopment Analysis (DEA) for Nutritional Efficiency Modeling — Jan 2026 – Aug 2026
Research Assistant, Stony Brook University | Advisors: Prof. Herbert Lewis and Prof. Thomas Sexton

- Used USDA nutrition data and DEA to score the relative nutritional efficiency of foods based on beneficial nutrients and dietary risk factors, with potential applications in institutional meal planning.
- Constructed binary and continuous DEA models, which produced different efficiency scores and rankings. Traced anomalous early results to nutrient selection and model constraints to validate outputs.
- Decomposed inefficiency into food-level and nutrient-level factor efficiencies to distinguish whether a food's efficiency score was driven by the food as a whole or by specific nutrients.
- Moved early Excel, VBA, and Solver models into a Python script that automates optimization across food groups, improving reproducibility.

Professional Experience

Stony Brook University — Stony Brook, NY
Marketing and Website Management Graduate Assistant — Jun 2025 – Present

- Build and maintain Graduate School web pages using HTML/CSS and JavaScript, implementing structural, accessibility, and content updates.
- Write and edit university news stories, program communication, and digital content, distilling complex academic and research information into clear materials for students, faculty, staff, and external audiences.

Professional Experience (continued)

Renewable Energy Long Island [Live dashboard] Brentwood, NY
Data Visualization Intern May 2026 – Aug 2026

- Built and published an interactive clean-energy dashboard, as part of a pilot with NYSERDA, the Long Island Clean Energy Hub, and Cornell Cooperative Extension, comparing 15 Long Island municipalities across community designation and renewable programs. Reviewed by Cornell Cooperative Extension of Suffolk County.
- Cleaned, validated, and standardized survey responses from disparate public sources and prior interns across all 15 municipalities, making judgments about which responses were meaningful for cross-municipality comparison.
- Worked with stakeholders to define performance indicators and design dashboard features aligned with municipal reporting, comparison, and planning needs.

Vivvi Early Learning New York, NY
Admissions Intern Jun 2022 – Aug 2022

- Analyzed enrollment workflows and client data to identify process bottlenecks, contributing to a 39% increase in enrollment and improvements in the prospective-family onboarding process.
- Automated administrative workflows using Google Apps Script and maintained enrollment records for 240+ families, improving data consistency and reducing repetitive manual work.

Teaching Experience

Stony Brook University Stony Brook, NY

MBA 506: Leadership, Team Effectiveness, and Communications Jan 2026 – May 2026
Teaching Assistant

- Evaluated graduate-level written assignments and provided detailed feedback on argumentation, organization, evidence, and clarity, helping students strengthen both their analytical reasoning and professional writing.
- Assessed students' application of leadership, decision-making, and organizational behavior concepts while maintaining consistent grading standards across written coursework.

WSE 105: Opportunities in STEM and Beyond Jan 2024 – May 2024
Teaching Assistant

- Mentored students in the Women in Science and Engineering Honors Program through classroom facilitation, academic support, grading, and individual student guidance.
- Collaborated with faculty to foster an inclusive learning environment designed to support academic success, community building, and participation in STEM.

AMS 210: Linear Algebra Aug 2022 – Dec 2022
Teaching Assistant

- Supported instruction for 170+ students through weekly office hours, individualized problem-solving support, and responses to student questions.

Technical Skills

Programming	Python (pandas, NumPy, scikit-learn, Matplotlib, PyTorch), R, SQL, VBA, Java, JavaScript, TypeScript, C
Quantitative Methods	Mathematical optimization, data envelopment analysis (DEA), multi-criteria decision analysis (MCDA, including TOPSIS), hierarchical clustering, regression, time-series analysis, hypothesis testing, model validation
Web Development	HTML/CSS, React, Node.js, Express.js, Electron.js, Vite
Databases	MySQL, PostgreSQL, MongoDB
Tools	Excel (Solver, OpenSolver), Google Sheets, Jupyter Notebook, LaTeX, Git, VS Code